Classical Mechanics and Dynamical Systems

Lecture Notes

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(transcribed from handwritten notes by Vishal Rao)

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1 Lecture 13: Dynamical Symmetry – 1

Understanding the relation of symmetry $\iff$ invariance $\Rightarrow$ conservation laws. The first step in this regard was taken by Emmy Noether.

$$\text{(conserved quantities: } P_i,\ L,\ E(H))$$

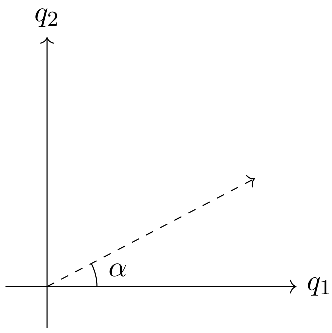
1.1 Noether's Theorem

Let us take a case of a particle moving in 2-D:

$$H(q_1, q_2, p_1, p_2) = \frac{\vec{p}^2}{2m} + V(q_1, q_2), \qquad \vec{p}^2 = p_1^2 + p_2^2$$

If the potential has spherical symmetry (in 2-D its rotational) then we know that H is a constant of motion (C.O.M.).

$$\text{if} \quad V(q_1, q_2) = f(r), \qquad r = \sqrt{q_1^2 + q_2^2}$$



In the rotated frame of reference, what remains constant:

1. H
2. Equations of motion (EOM) don't change
3. Solution space/set do not change

Q) Does the solution remain the same (constant)?

Let's take the simplest case:

$$\begin{aligned} x^2 &= 4 \\ x^2 - 4 &= 0 \\ (x-2)(x+2) &= 0 \\ x &= 2, -2 \qquad \{-2, 2\} \end{aligned}$$

After a rotation transformation $x' = -x$:

$$\begin{aligned} x'^2 &= 4 \\ (x'-2)(x'+2) &= 0 \\ x' &= -2, +2 \qquad \{-2, 2\} \end{aligned}$$

Since $\underline{x' = -x}$, we deduce that the $+2$ root changed to -2 and $-2 \to +2$, but the set of solutions remains constant/same, even though the elements of the set of solutions can convert into one another.

1.2 Dynamical Symmetry

So the definition of dynamical symmetry becomes:

Dynamical Symmetry: The dynamical symmetry of a given system is a set of transformations of phase space variables, such that the EOM don't change, and as a consequence, the set of solutions do not change.

1.3 Understanding Dynamical Symmetry in the Lagrangian Formalism

$$L(q,\dot{q}) \longrightarrow L'(Q(\kappa),\dot{Q}(\kappa)), \qquad \kappa = \text{transformation parameter}$$

(Autonomous dynamical system) $\to L'$, with $L' = L$ if $\kappa = 0$ (by definition), and $Q(0) = q$.

EOM do not change if $L' = L$, or, Invariance $\Rightarrow L' = L$.

$$\frac{dL(Q(\kappa),\dot{Q}(\kappa))}{d\kappa} = 0$$

$$\Rightarrow \frac{\partial L}{\partial Q}\frac{dQ}{d\kappa} + \frac{\partial L}{\partial \dot{Q}}\frac{d\dot{Q}}{d\kappa} = 0$$

$$\Rightarrow \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{Q}}\right)\frac{dQ}{d\kappa} + \frac{\partial L}{\partial \dot{Q}}\frac{d}{d\kappa}\left(\frac{dQ}{dt}\right) = 0$$

κ, t are independent, so

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{Q}}\right)\frac{dQ}{d\kappa} + \frac{\partial L}{\partial \dot{Q}}\frac{d}{dt}\left(\frac{dQ}{d\kappa}\right) = 0$$

$$\Rightarrow \frac{d}{dt}\left(\left(\frac{\partial L}{\partial \dot{Q}}\right)\left(\frac{dQ}{d\kappa}\right)\right) = 0$$

$$\Rightarrow \frac{\partial L}{\partial \dot{Q}}\left(\frac{dQ}{d\kappa}\right) = \text{const. of motion}$$

In our frame,

$$\frac{\partial L}{\partial \dot{Q}}\left(\frac{dQ}{d\kappa}\right) = \frac{\partial L}{\partial \dot{q}}\left(\frac{dq}{d\kappa}\right)\Bigg|_{\kappa=0} = p\,\frac{dq}{d\kappa}\Bigg|_{\kappa=0} = \text{const}$$

Remember, that the Lagrangian (L) is not unique: we can add the total time derivative of some function of q, t; therefore it is not necessary that

$$\text{Invariance} \Rightarrow L' = L$$

We can have $L' \neq L$ and still EOM don't change.

Note: The crucial point of Noether's theorem is that the set of transformations should be continuous – that's why we differentiate w.r.t. κ – and it must be connected with the identity (original coordinates $(q,\dot{q})$). In other words, if we have the original coordinates, then we make a transformation continuously starting from no-transformation at all. Parity is *not* a continuous transformation.

Continuous Transformations:

- Rotation
- Shift in origin
- Shift in origin in time
- Shear, scale
- Gauge transformations (vary κ continuously) $\to$ (gauge invariance $\Rightarrow$ leads to conservation of charge).

Symmetry: If the Lagrangian is invariant under a continuous set of transformations, then Noether's theorem tells us there exists a conserved quantity for each transformation parameter.

In shift of origin we have $(x, y, z) \to (x+\alpha, y+\beta, z+\gamma)$; we have 3 parameters of transformation, i.e. α, β, γ. So, according to Noether's theorem, we will have 3 conserved quantities.

1.4 Understanding Dynamical Symmetry in the Hamiltonian Framework

$$H(q,p): \qquad \dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i} \qquad \boxed{\dot{X} = J\nabla H} \text{ (set of EOM)}$$
$$\{q_i, q_j\} = 0 = \{p_i, p_j\}, \qquad \{q_i, p_j\} = \delta_{ij}$$

Q. What set of transformations leaves Hamilton's EOM unchanged?

Ans. Canonical transformations.

$$\dot{Q}_i = \frac{\partial K}{\partial P_i}, \qquad \dot{P}_i = -\frac{\partial K}{\partial Q_i} \qquad \boxed{\dot{\xi} = J\nabla_\xi K} \text{ (EOM)}$$

also

$$\{Q_i, P_j\} = \delta_{ij}, \qquad \{Q_i, Q_j\} = 0 = \{P_i, P_j\}$$

To see the set of solutions remain unchanged, we need only those canonical transformations for which $K = H$; else it won't be a symmetry (How?).

Suppose $X = (q,p) \xrightarrow{\text{C.T.}} \xi = (Q, P)$, where X, ξ are sets of variables.

1.5 Symplectic Matrices and Canonical Transformations

$$\{Q_i, P_j\} = \delta_{ij} = \sum_{k=1}^{n} \left(\frac{\partial Q_i}{\partial q_k}\frac{\partial P_j}{\partial p_k} - \frac{\partial Q_i}{\partial p_k}\frac{\partial P_j}{\partial q_k} \right)$$
$$\{Q_i, Q_j\} = 0 = \sum_{k=1}^{n} \left(\frac{\partial Q_i}{\partial q_k}\frac{\partial Q_j}{\partial p_k} - \frac{\partial Q_i}{\partial p_k}\frac{\partial Q_j}{\partial q_k} \right)$$
$$\{P_i, P_j\} = 0 = \sum_{k=1}^{n} \left(\frac{\partial P_i}{\partial q_k}\frac{\partial P_j}{\partial p_k} - \frac{\partial P_i}{\partial p_k}\frac{\partial P_j}{\partial q_k} \right)$$
$$\Rightarrow \left(\frac{\partial \xi}{\partial x}\right)^T J \left(\frac{\partial \xi}{\partial x}\right) = J$$

All three relations must be valid to qualify a transformation as canonical.

$$J = \begin{pmatrix} O_n & I_n \\ -I_n & O_n \end{pmatrix}, \qquad I_n, O_n \text{ are } n \times n \text{ matrices}$$

(Margin note: the transformation matrix $M = \partial\xi/\partial x$ is $2n \times 2n$, i.e. $q = 2n,\ p = 2n$.)

For any $\{A, B\}$ can be written as $\{A, B\} = (\nabla A)^T J (\nabla B)$.

If a matrix M follows

$$M^T J M = J,$$

then M is called a symplectic matrix.

Recall, $J^2 = -I$, $J^T = J^{-1} = -J$.

So symplectic matrices have inverses, so canonical transformations' matrix is invertible: $M = \left(\dfrac{\partial\xi}{\partial x}\right)$ is non-singular, $|M| \neq 0$.

We know canonical transformations must be invertible transformations. Also, the product of two symplectic matrices is also a symplectic matrix (as symplectic matrices form a group).

$$M_1, M_2, M_3 \to \text{symplectic matrices}, \qquad M_1 \cdot M_2 = M_3$$

Physically it implies that two canonical transformations applied serially are equivalent to a single canonical transformation.

Note: The inverse-composition law suggests the set of canonical transformations for an n-dof Hamiltonian system forms a group. It is the group of $2n \times 2n$ symplectic matrices, called

$$Sp(2n, \mathbb{R}) \hookrightarrow \text{Group of C.T.}$$

Note: J is also a symplectic matrix.

$$M^T J M = J \quad \Rightarrow \quad \text{LHS} = J^T J J = (-J)J^2 = (-J)(-I) = J = \text{RHS}$$

Q) What is the C.T. corresponding to symplectic matrix J?

$$Q = -p, \quad P = q \qquad (M = J)$$

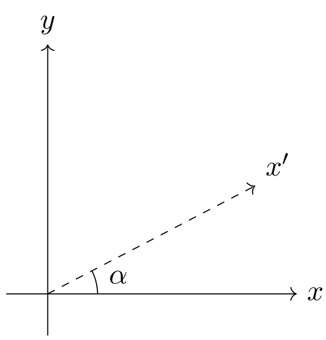
1.6 Defining Rotation

Note: We know that the group of rotations in 3 dimensions has three parameters (for rotation),

$$SO(n) \hookrightarrow \text{Group of rotations in } n\text{-dimensions with orthogonal matrices, determinant } 1$$

(Digression:) **Q.** How many parameters for n-D?

Ans. (a) Rotation in 2-D about the point (origin).



$$SO(2) : \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix} \quad \text{orthogonal}$$

Set of transformation is homogeneous (origin unchanged). 2-D rotation (linear, homogeneous) is $SO(2)$, $|SO(2)| = +1$, length unchanged.

Defining rotation (for n-dimensions):

- Linear
- Homogeneous
- Distances unchanged $\Rightarrow$ matrix should be orthogonal
- $|SO(n)| = 1$ (unimodular)

Note: In 3-D, if we rotate by a single axis, it is not termed as "rotation," as from the definition we have not defined rotation as rotation about a single axis. For rotation T should be linear, homogeneous, orthogonal, and unimodular.

Note: If the transformation is homogeneous (origin fixed) and to achieve distances unchanged, the transformation matrix should be orthogonal; and if we want orientation to be unchanged we need it to be unimodular, $|M| = 1$. So, as we know det(orthogonal M) $= \pm 1$, we choose $+1$ for rotation. For reflection $|M| = -1$.

In 3-D we can do rotation on a plane, where two variables are affected at a time. For two variables affected at a time, the number of rotations is ${}^nC_2$ for n-dimensions.

$$\text{no. of rotations} \Rightarrow {}^nC_2 = \frac{n(n-1)}{2}$$

for $n = 3$: no. of rotations $= \dfrac{3 \cdot 2}{2} = 3$.

So we think (mistakenly) that rotation is occurring about an axis, but really we mean rotation in the xy plane, yz plane, xz plane.

Hence, only for 3-D is $\dfrac{n(n-1)}{2} = n$.

1.7 Generators of Transformation Groups

So $SO(n)$ has ${}^nC_2$ generators.

In exactly the same way, we can ask how many generators does the symplectic group $Sp(2n, \mathbb{R})$ have? (Or) How many possible parameters do we need to specify all possible canonical transformations. Let us make an infinitesimal transformation.

$$M = I + \xi\, G_{(2n\times 2n)}, \qquad G: \text{ Generator of transformation matrix}, \quad \xi: \text{ parameter}$$

$$\hookrightarrow M^T J M = J$$

$$\Rightarrow (I+\xi G)^T J(I+\xi G) = J$$
$$\Rightarrow (I+\xi G^T)\cdot J\cdot(I+\xi G) = J$$
$$\Rightarrow (J+\xi G^T J)(I+\xi G) = J$$
$$\Rightarrow J+\xi G^T J+\xi JG+\xi^2 G^T JG = J \qquad \text{(dropping } \xi^2\text{)}$$
$$\Rightarrow J+\xi G^T J+\xi JG = J$$
$$\Rightarrow \boxed{G^T J = -JG}$$
$$\Rightarrow G^T = -JGJ^{-1} \qquad (J^{-1} = -J)$$
$$\Rightarrow \boxed{G^T = JGJ}$$

So the generator of a symplectic transformation must satisfy the above equation. Symplectic matrices are generated by those matrices which satisfy $G^T = JGJ$.

$$\boxed{\text{Exercise} \Rightarrow \text{Show that } Sp(2n) \text{ has } 2(2n+1) \text{ generators.}}$$

What are generators? The way we look at it can be interpreted by looking at an infinitesimal transformation.

$$SO(2) = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix} \xrightarrow{\alpha\to\delta\alpha} \begin{pmatrix} 1 & \delta\alpha \\ -\delta\alpha & 1 \end{pmatrix} = I + \begin{pmatrix} 0 & \delta\alpha \\ -\delta\alpha & 0 \end{pmatrix} = I + \delta\alpha \underbrace{\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}}_{\text{Generator}}$$

with $\delta\alpha$ the parameter.

For $SO(3)$ we will have 3 such generators, and for $Sp(2n)$ we will have $2(2n+1)$ generators.

Note: We actually like to write generators as Hermitian matrices, because we need most of the time to exponentiate a Hermitian matrix, as e^{iH} = unitary matrix.

1.8 The 2-D Isotropic Oscillator

$$H = \frac{1}{2}(q_1^2 + p_1^2 + \omega^2 q_2^2 + p_2^2) \quad \text{quasi periodic } (\omega = \text{irrational})$$

periodic but complicated (ω=rational).

If ($\omega = 1$):

$$H = \frac{1}{2}(q_1^2 + p_1^2 + q_2^2 + p_2^2)$$

isotropic (same spring constant k in all directions).

H is invariant under $SO(4) \hookrightarrow$ set of rotations in q-p phase space.

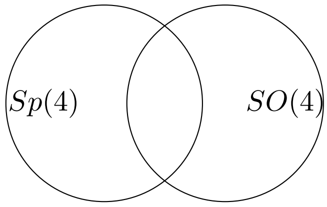
Note: We could have dynamical symmetry arising as a consequence of transformations not just of potentials (not just of physical coordinates), but it could also involve the momenta (phase-space).

$$\text{no. of generators } SO(4) = {}^4C_2 = 6 \qquad \text{—(1)}$$

set of canonical transformations ($n = 2$) $\to Sp(2{\times}2)$, $\quad Sp(4)$ has $2(4{+}1) = 10$ generators —(2)

From (1) and (2) we deduce that all transformations which leave the Hamiltonian unchanged need not be canonical transformations; similarly, all C.T. need not leave H unchanged.

For symmetry we need the transformation to be canonical *and* to leave H unchanged ($K = H$).



$$\to \text{Dynamical symmetry group is } Sp(4) \cap SO(4) \sim SU(2) \hookrightarrow \text{isomorphic}$$

$SU(2)$: set of 2×2 matrices which are unitary and unimodular ($|M| = 1$).

U is unitary $\Rightarrow$ $\boxed{U^\dagger U = I}$

Q. Is it true $U^\dagger U = I = UU^\dagger$?

Ans. Only true for finite-dimensional matrices (as the left inverse becomes equal to the right inverse). But not for infinitesimal matrices: $U^\dagger U = I \not\Rightarrow UU^\dagger$. But we are concerned with finite-dimensional matrices, so it does not matter.

$$U^\dagger U = I \Rightarrow \text{partial isometry}, \qquad UU^\dagger = I \Rightarrow \text{ ''}$$

Verify that the following J_1, J_2, J_3 are constants of motion:

$$J_2 = \frac{1}{2}(q_1 q_2 + p_1 p_2), \qquad J_1 = \frac{1}{4}(q_1^2 + p_1^2 - q_2^2 - p_2^2), \qquad J_3 = \frac{1}{2}(q_1 p_2 - q_2 p_1)$$

Verify: $\{J_i, H\} = 0$, $\{J_i, J_j\} = \epsilon_{ijk} J_k$.

2 Lecture 14: Finding the Dynamical Symmetry Group $SU(N)$

Recall the 2-D isotropic oscillator:

$$H = \frac{1}{2}(q_1^2 + p_1^2 + q_2^2 + p_2^2) \qquad (M = I,\ \omega = 1)$$

$$F_1 = H_1 = (q_1^2 + p_1^2), \qquad F_2 = H_2 = (q_2^2 + p_2^2)$$

It's 2-dof and we have 2 constants of motion which are in involution with each other ($\{F_1, F_2\} = 0$). (Integrable.)

<u>Note:</u> The dynamical symmetry will be that set of transformations of the 4 phase space variables which leaves H, EOM unchanged, and hence preserves the solution set.

$Sp(2n) = Sp(4) =$ Group of canonical transformations for this H
$SO(4) = SO(4) =$ Group of rotations which leave H unchanged
$SU(n) = SU(2) =$ Group of C.T. which leaves H unchanged (also called the dyn. symmetry of this H)

2.1 The Group $SU(2)$

$SU(2)$ consists of all 2×2 matrices $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, such that (implies)

$$M^\dagger M = I, \qquad \det(M) = +1$$

There are $4 \times 2 = 8$ parameters as each entry can be a complex number:

$$a = a_1 + ia_2, \quad b = b_1 + ib_2, \quad c = c_1 + ic_2, \quad d = d_1 + id_2 \qquad \hookrightarrow 8 \text{ parameters}$$

Using $M^\dagger M = I$ or $M^{-1} = M^\dagger$, we will have 4 conditions (one for each element). Therefore the number of parameters is reduced to 4. After imposing $|M| = +1$, one more parameter is lost. So a total of three parameters are left.

(Digression:)

$$M^\dagger = M^{-1}, \qquad M = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

$$M^\dagger = \begin{pmatrix} \alpha^* & \gamma^* \\ \beta^* & \delta^* \end{pmatrix} \qquad \text{—(1)}$$

$$M^{-1} = \frac{1}{|M|} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}^T = \frac{1}{|M|} \begin{pmatrix} \delta & -\gamma \\ -\beta & \alpha \end{pmatrix} \qquad \text{—(2)}$$

Equating (1) and (2), and $|M| = \alpha\delta - \beta\gamma = 1$:

$$\alpha^* = \delta \text{ —(3)}, \qquad \beta^* = -\gamma \text{ —(4)}, \qquad \gamma^* = -\beta \text{ —(5)}, \qquad \delta^* = \alpha \text{ —(6)}$$

From $\alpha^* = \delta$ and $\delta^* = \alpha$: $(\alpha_1 + i\alpha_2)^* = (\delta_1 + i\delta_2) \Rightarrow \alpha_1 = \delta_1$, and similarly $(\delta_1 + i\delta_2)^* = \alpha_1 + i\alpha_2 \Rightarrow$ **[?]** (showing that (3) and (6) are not independent conditions).

So M becomes

$$M = \begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix}, \qquad |M| = 1, \quad \alpha\delta - \beta\gamma = 1, \quad \alpha\alpha^* + \beta\beta^* = 1, \quad |\alpha|^2 + |\beta|^2 = 1$$

Hence, any 2×2 unimodular, unitary matrix can be written as

$$\begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix} \quad \text{with} \quad |\alpha|^2 + |\beta|^2 = 1, \qquad \alpha = \alpha_1 + i\alpha_2, \quad \beta = \beta_1 + i\beta_2$$

The generators of $SU(2)$ are constants of motion which are *not* in involution with each other, and it turns out these are J_1, J_2, J_3:

$$J_1 = \frac{1}{4}(q_1^2 + p_1^2 - q_2^2 - p_2^2), \qquad J_2 = \frac{1}{2}(q_1 q_2 + p_1 p_2), \qquad J_3 = \frac{1}{2}(q_1 p_2 - p_1 q_2)$$

$$\{J_1, J_2\} = J_3, \qquad \{J_2, J_3\} = J_1, \qquad \{J_3, J_1\} = J_2 \qquad \text{(or) } \{J_i, J_j\} = \epsilon_{ijk} J_k$$

(So 3 parameters require 3 generators.)

2.2 Counting Independent Parameters of $SU(n)$

Q) How many independent parameters in an $(n \times n)$ unitary matrix?

Ans. $2n^2 - n^2 = n^2$.

So for $SU(n)$ we have $(n^2 - 1)$ independent parameters, where 1 parameter got reduced from the $|M| = \pm 1$ condition.

Therefore, e.g. $SU(3)$ has $3^2 - 1 = 8$ generators.

For the 3-D isotropic oscillator:

$Sp(2n)$: $Sp(6)$ is the group of C.T.
$SO(N)$: $SO(6)$ is the group of symmetry transformations
$SU(3)$ is our required dynamical symmetry group.

(Margin note: dimension of phase space.)

2.3 The Kepler Problem

$$H(\vec{r},\vec{p}) = \frac{\vec{p}^2}{2m} + \left(-\frac{k}{r}\right) \qquad \text{(assuming interaction is attractive)}$$

Note:

$V(r) = -\dfrac{k}{r}$ and $V(r) = r^2$ *are the only potentials for which bounded orbits are closed curves* (periodic orbits).

$V(r) = r^2$: 3-D oscillator, with $SU(3)$.

Q) What is the symmetry group of Kepler's Hamiltonian?

C.O.M.: $H,\ \vec{L}^2,\ \vec{L}\cdot\hat{n} \to 3$ independent constants of motion which are in involution with each other.

$\vec{L}$ is a C.O.M. but $\{L_i, L_j\} = \epsilon_{ijk}L_k$ (not in involution with each other).

Note: Any central potential problem will have these C.O.M., but there is something special with the Kepler problem, as there exists a further C.O.M., $\vec{A}$:

$$\vec{A} = \vec{p}\times\vec{L} - \frac{mk\vec{r}}{r} \qquad (\vec{A} = f(L),\ \text{so it's not independent})$$

(Laplace–Runge–Lenz vector.)

Verifying $\vec{A}$ is a C.O.M.:

$$\frac{d\vec{A}}{dt} = \frac{d\vec{p}}{dt}\times\vec{L} + \vec{p}\times\frac{d\vec{L}}{dt} + \frac{mk}{r^2}\frac{dr}{dt}\vec{r} - \frac{mk}{r}\frac{d\vec{r}}{dt}$$

for $V = -k/r$:
$F = -\nabla V = -\nabla(-k/r) = \nabla(k/r) = -k/r^2\,\hat{e}_r$
$\dfrac{d\vec{p}}{dt} = F = -\dfrac{k}{r^3}\vec{r}$

(since $\vec{L}$ is a C.O.M., $d\vec{L}/dt = 0$.)

$$\frac{d\vec{A}}{dt} = -\frac{mk}{r^3}\vec{r}\times(\vec{r}\times\vec{p}) + \frac{mk}{r^2}\frac{dr}{dt}\vec{r} - \frac{mk}{r}\frac{d\hat{r}}{dt}$$

$$\frac{d\vec{A}}{dt} = -\frac{k}{r^3}\big(\vec{r}\times(\vec{r}\times\vec{p})\big) + \frac{mk}{r^2}\frac{du}{dt}\vec{r} - \frac{k\vec{p}}{r} \qquad (\vec{p} = m\dot{\vec{r}})$$

$$\boxed{\vec{a}\times(\vec{b}\times\vec{c}) = \text{vector in the plane of } \vec{b},\vec{c} = \alpha\vec{b} + \beta\vec{c}}$$

LHS is linear; α,β are scalars formed out of a,b,c. As LHS is linear, RHS must be linear:

$$\Rightarrow \alpha = \lambda(\vec{a}\cdot\vec{c}), \qquad \beta = \mu(\vec{a}\cdot\vec{b})$$

$$\vec{a}\times(\vec{b}\times\vec{c}) = \lambda(\vec{a}\cdot\vec{c})\vec{b} + \mu(\vec{a}\cdot\vec{b})\vec{c}$$

(λ and μ should be universal; they can't depend on $\vec{a},\vec{b},\vec{c}$.)

Interchanging b,c: $\vec{a}\times(\vec{c}\times\vec{b}) = -\vec{a}\times(\vec{b}\times\vec{c})$. The only way this can happen is $\mu = -\lambda$:

$$\vec{a}\times(\vec{b}\times\vec{c}) = \lambda\big[(\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}\big]$$

It turns out $\lambda = 1$, using the special case $\hat{\imath} \times (\hat{\jmath} \times \hat{\jmath}) = \cdots\ \hat{\imath} \times \hat{k} = -\hat{\jmath}$:

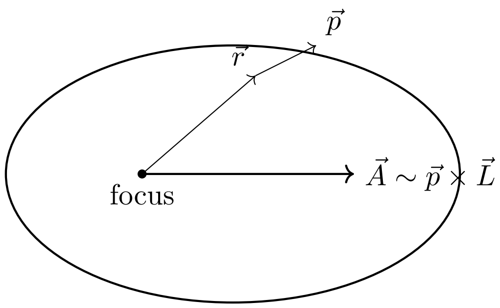
$$\lambda\big[(\hat{\imath}\cdot\hat{\jmath})\hat{\imath} - (\hat{\imath}\cdot\hat{\imath})\hat{\jmath}\big] = -\hat{\jmath} \ \Rightarrow\ \boxed{\lambda = 1}$$

$$\begin{aligned}
\frac{d\vec{A}}{dt} &= -\frac{k}{r^3}\left[(\vec{r}\cdot\vec{p})\vec{r} - r^2\vec{p}\right] + \frac{mk}{r^2}\frac{du}{dt}\vec{r} - \frac{k\vec{p}}{r} \\
&= -\frac{k}{r^3}(\vec{r}\cdot\vec{p})\vec{r} + \frac{k}{r}\vec{p} + \frac{mk}{r^2}\frac{du}{dt}\vec{r} - \frac{k\vec{p}}{r} \\
&= -\frac{mk}{r^3}\left(\vec{r}\cdot\frac{d\vec{r}}{dt}\right)\vec{r} + \frac{mk}{r^2}\frac{du}{dt}\vec{r} \qquad \left(\vec{u}\cdot\frac{d\vec{u}}{dt} = u\frac{du}{dt}\right) \\
&= -\frac{mk}{r^3}\left(r\frac{du}{dt}\right)\vec{r} + \frac{mk}{r^2}\frac{du}{dt}\vec{r}
\end{aligned}$$

$$\frac{d\vec{A}}{dt} = 0 \qquad \text{therefore } \vec{A} \text{ is a C.O.M.}$$

$\vec{A}$ is a C.O.M., therefore it remains unchanged in direction and magnitude.

$$\vec{A} = \vec{p} \times (\vec{r} \times \vec{p}) - \frac{mk}{r}\vec{r}$$



(direction of $\vec{A}$ lies along the semi-major axis.)

The $d\vec{A}/dt = 0$ implies the orbit does not precess.

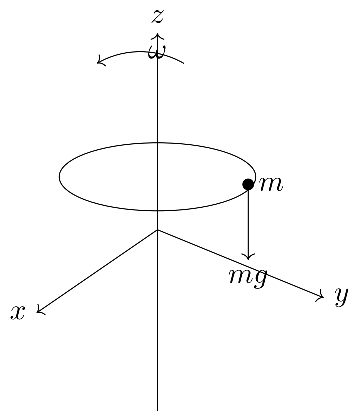
- Binary stars precess $\approx 4^\circ$ per orbit
- Precession of Mercury $\approx 500''$ per century

So in total we got $H, \vec{L}, \vec{A}$, a total of $(1 + 3 + 3) = 7$ constants of motion, but phase space is only 6-dimensional. It turns out that they are not independent of each other (A^2 is expressed in terms of H and L^2; $\vec{A}\cdot\vec{L} = 0$).

Note: $d\vec{A}/dt = 0$ independent of the sign of k, so $\vec{A}$ exists for repulsive forces also.

- The dynamical symmetry group is $SU(4)$ instead of $SU(3)$. This extra symmetry is seen in terms of degeneracy in quantum mechanics. This extra symmetry is because of $\vec{A}$.

2.4 Problem 1: Bead on a Rotating Hoop of Wire



Bead on a rotating hoop of wire.

$$V = \begin{cases} 0 & \text{at } z = 0 \\ mgz & \text{at } z = z \end{cases}$$

$$L = T - V, \qquad L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz$$

Constraint: $y^2 + (z - R)^2 = R^2$.

The moment we provided ω to the hoop, we need cylindrical symmetry (s, ϕ, z):

$$x = s\cos\phi, \quad \dot{x} = \dot{s}\cos\phi - s\sin\phi\,\dot{\phi}$$

$$y = s\sin\phi, \quad \dot{y} = \dot{s}\sin\phi + s\cos\phi\,\dot{\phi}$$

$$z = z, \quad \dot{z} = \dot{z}$$

So L becomes:

$$L = \frac{1}{2}m\big[\dot{s}^2 + s^2\dot{\phi}^2 + \dot{z}^2\big] - mgz \qquad \text{—(1)}$$

Constraint becomes:

$$(x^2 + y^2) + (z - R)^2 = R^2 \;\Rightarrow\; s^2 + (z - R)^2 = R^2 \qquad \text{—(2)}$$

also, $\dot{\phi} = \omega$ —(3)

Eq. (2) implies $z = R \pm \sqrt{R^2 - s^2}$; we should choose $z = R - \sqrt{R^2 - s^2}$, as for $s = 0,\; z \to 0$.

$$\dot{z} = \frac{\pm(2s)\dot{s}}{2\sqrt{R^2 - s^2}} = \frac{s\dot{s}}{\sqrt{R^2 - s^2}} \qquad \text{—(4)}$$

Using eqs. (3),(4) in (1), L becomes:

$$L = \frac{m}{2}\left[\dot{s}^2 + s^2\omega^2 + \frac{s^2\dot{s}^2}{R^2 - s^2}\right] - mg\big(R - \sqrt{R^2 - s^2}\big)$$

$$= \frac{m}{2}\left[\frac{\dot{s}^2R^2 - \dot{s}^2s^2 + s^2\omega^2R^2 - s^2\omega^2s^2 + s^2\dot{s}^2}{R^2 - s^2}\right] - mgR + mg\sqrt{R^2 - s^2}$$

$$L(s, \dot{s}) = \frac{m}{2}\left(\frac{\dot{s}^2R^2}{R^2 - s^2}\right) + \frac{1}{2}ms^2\omega^2 - mgR + mg\sqrt{R^2 - s^2}$$

Since L only depends on one dynamical variable of space (s), it has only one degree of freedom (s).

$$\boxed{\frac{\partial L}{\partial s} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{s}}\right)}$$

$$p_s = \frac{\partial L}{\partial \dot{s}} = \frac{mR^2 \dot{s}}{R^2 - s^2} \;\Rightarrow\; \dot{s} = \frac{p(R^2 - s^2)}{mR^2}$$

$$\begin{aligned} H(s,p) &= p\dot{s} - L \\ &= p\left[\frac{p(R^2 - s^2)}{mR^2}\right] - \frac{m}{2}\left[\frac{p(R^2 - s^2)}{mR^2}\right]^2 \frac{R^2}{R^2 - s^2} - \frac{1}{2}m\omega^2 s^2 + mgR - mg\sqrt{R^2 - s^2} \end{aligned}$$

$$H(s,p) = \frac{p^2(R^2 - s^2)}{mR^2} - \frac{1}{2}m\omega^2 s^2 + mgR - mg\sqrt{R^2 - s^2}$$

$$\dot{s} = \frac{\partial H}{\partial p} = \frac{2p(R^2 - s^2)}{mR^2}$$

$$\dot{p} = -\frac{\partial H}{\partial s} = \frac{+2p^2 s}{mR^2} + m\omega^2 s + \frac{mgs}{\sqrt{R^2 - s^2}}$$

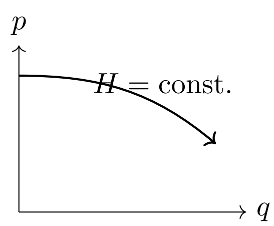
3 Lecture 15: Randomness in phase-space

How do we find the equation of the phase-trajectory using Hamilton's equations of motion? $H(q,p)$

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Ans: we have to solve these equations; if these equations are integrable.

For 1-d.o.f, C.O.M $= H$ (C.O.M $=$ constants of motion), so the trajectory is just given by $H = $ const.



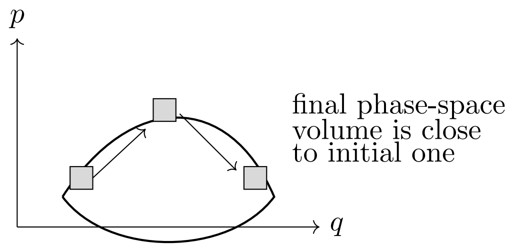
For n-d.o.f, we need n constants of motion (C.O.M) and find the canonical transformation (C.T.) which takes us to (I, θ) variables, and this task is not easy to do.

–

(If a system is integrable, we can predict what will happen at any time t in the future, provided the initial phase-space point.)

Integrable $\longrightarrow$ Periodic motion
– quasi-periodic
– ergodic motion
– mixing / instability on the average
– exponential [instability]   arise from lack of enough C.O.M.

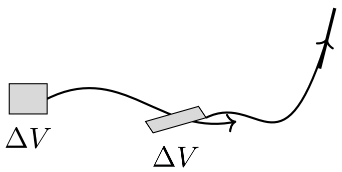
(increasing complexity, top to bottom)



Ergodicity

If the initial volume element in phase space visits the neighbourhood of every point of (part of) phase-space, in a given sufficient time, then we say the motion is ergodic.

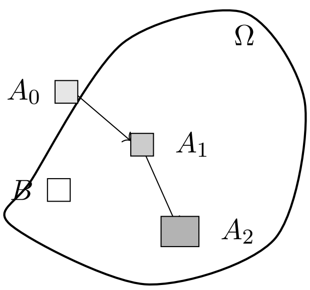
Note:- Worse can happen to the initial volume element, even when the volume itself is preserved.



As time goes on, such pieces of the volume element can go as far as the size of the system (arbitrarily far); such kind of motion is ergodic too, but there is a special name for this motion, called mixing.

Mixing

Suppose the volume of the full phase space is Ω.



A_0 – initial phase-space volume; $\mu \to$ measure.

If A_0 is completely mixed up, how much of A_0 is present in B?

$$\frac{\mu(A_n \cap B)}{\mu(A_0)} = \frac{\mu(B)}{\mu(\Omega)}$$

if,

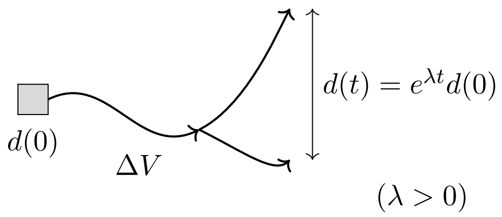
$$\underset{n\to\infty}{\mathrm{Lt}}\, \mu(A_n \cap B) = \frac{\mu(A_0)\,\mu(B)}{\mu(\Omega)}$$

then the dynamics is said to be strongly mixing.

Note:- Mixing implies ergodicity, whereas ergodicity does not imply mixing, as we can have ergodic motion without mixing (distortion of the phase-space element).

Rate of mixing

Typically, mixing occurs exponentially fast.

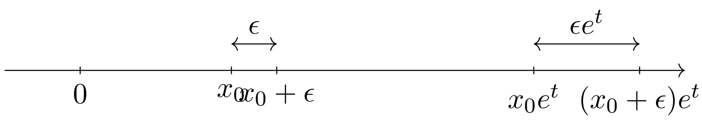


System is said to be in exponential sensitivity to initial conditions.

Example (1-D phase space) $\to$ hypothetical:

$$\dot{x} = x, \qquad x(t) = x(0)e^t$$

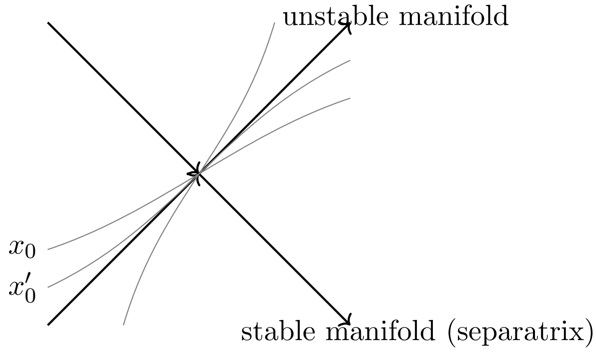
the initial separation is exponentially increasing.



the initial sep is exponentially increasing.

The analogy is not chaos

This is just an analogy to understand, but keep in mind $\dot{x} = x$ is integrable and this is *not* chaos.



We can see that x_0 decides the future of two small phase-space points that are very different, which are [nonetheless] in the neighbourhood [of each other]. So it is like [having] a separatrix at every point – that would imply chaos.

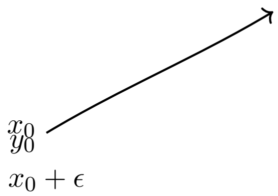
Conditions for having chaos

1. Bounded phase space.
2. Exponential sensitivity to initial conditions.
3. A dense set of unstable periodic orbits.

Exponential sensitivity in bounded phase space

If we have two different initial conditions x_0, y_0 with different phase trajectories, find the separation between $x(t), y(t)$.

$$|x_0 - y_0| = \epsilon$$



$$\lim_{t\to\infty} \frac{1}{t} \ln \frac{|x(t) - y(t)|}{|x(0) - y(0)|} = 0$$

as $t \to \infty$,

$$|x(t) - y(t)| \leq L \quad \text{(system size, bounded)}$$

To avoid such situations let us take $\epsilon \to 0$:

$$\lim_{\epsilon\to 0} \lim_{t\to\infty} \frac{1}{t} \ln \frac{|x(t) - y(t)|}{|x(0) - y(0)|} = 0, \qquad |x(0) - y(0)| = \epsilon$$

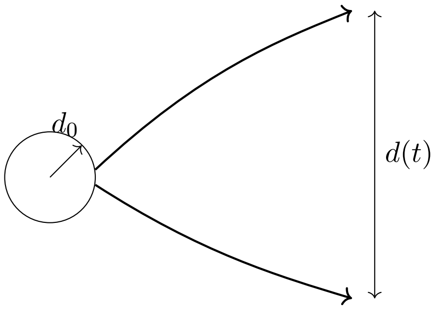
but let's invert the limits,

$$\lim_{t\to\infty} \lim_{\epsilon\to 0} \frac{1}{t} \ln \frac{|x(t) - y(t)|}{|x(0) - y(0)|} = \lambda(x_0) \qquad \text{— (1)}$$

"Liapunov exponent"

For n-dimensional phase space we need n Liapunov exponents, so there is a full spectrum of Liapunov exponents.

Equation (1) provides the *maximum* Liapunov exponent, as if a circle shrinks to a line.



(d_0 has exponential increment [along the unstable direction] while d_0 [along the other directions] has decreased with time.)

In continuous time dynamics, $N = 3$ is the minimum required for chaos. For discrete time dynamics, $N = 1$ will do.

$N =$ d.o.f. Particle

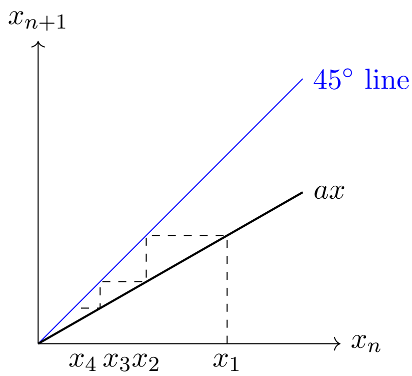
Instead of differential equations we call them difference equations; difference equations are called maps.

One-dimensional maps

$$x_{n+1} = f(x_n), \qquad n = 0, 1, 2, \ldots \quad \text{discrete time}$$

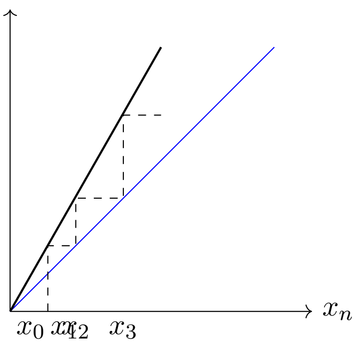
$(\Delta x_n = x_{n+1} - x_n = a)$

(i) $(a < 1)$: $\quad x_{n+1} = ax_n, \quad$ suppose $a < 1$.



So, $x = 0$ is a stable fixed point, as we tend to $x = 0$ from both sides.

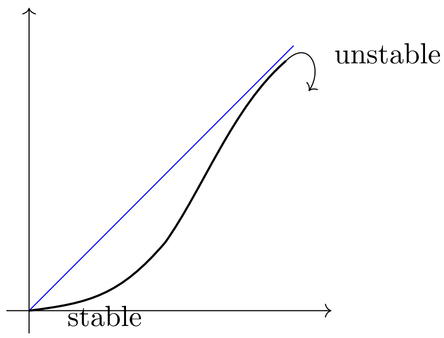
(ii) $(a > 1)$



So, $x = 0$ in this case is an unstable fixed point.

(iii) $a = 1$: map is $x = x$ (identity map) (trivial, degenerate case). Wherever the particle is, it does not move at all.

(iv) For any arbitrary map:



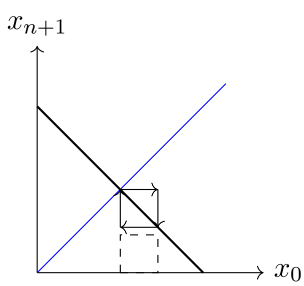
We can also see that the slope at a stable point is < 1 and the slope at an unstable fixed point is > 1.

So, if x^* is a fixed point then,

$$|f'(x^*)| > 1 \implies x^* \text{ is unstable}$$

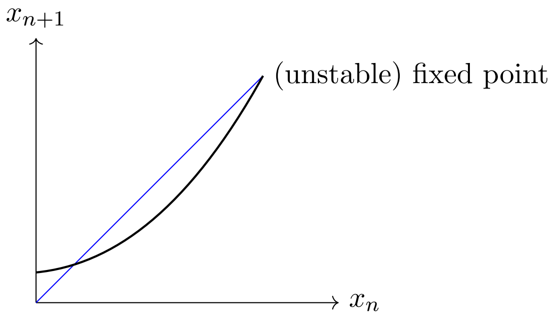
$$|f'(x^*)| < 1 \implies x^* \text{ is stable}$$

For $f'(x^*) = -1$: cyclic (loop).

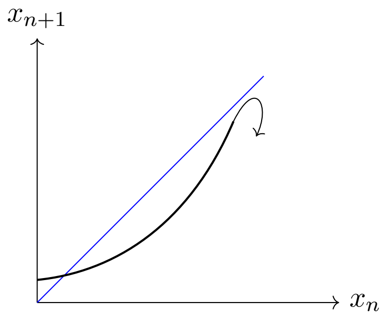


x^* = marginal fixed point (indifferent fixed point).

(v) Tangent case:-



(vi) (fixed point is missed): (Intermittency)



(Irregularity followed by long bursts of regularity.)

4 Lecture 16

Exponential divergence of phase trajectory can be characterised by the "Liapunov exponent".

Note:- We can find a finite-time "Liapunov exponent" but it is not relevant for the purpose we have in mind.

So the reason why we need a very long time Liapunov exponent is the following. If we start with variable x_0,

$$x_0 \longrightarrow x_1 \longrightarrow x_2 \longrightarrow \cdots \longrightarrow x_n \qquad (x_n = x_0 \text{ after } n \text{ time steps})$$

x_n is not computable, as the error is exponentially increasing (x_0, x_n).

Ex.: if we write the number in binary, and let's say the error doubles each step, then the error from the 100^{th} decimal place comes to the first decimal place after 100 steps.

Time average vs. ensemble average

If x represents any physical quantity, there may be some function of x, $\phi(x)$, whose average can be written as $\langle \phi(x) \rangle$.

$$\langle \phi(x) \rangle = \frac{1}{n} \sum_{j=0}^{n-1} \phi(x_j), \qquad n \to \text{time}$$

$\langle \phi(x) \rangle$ = time average.

$\phi(x_0)$	$\phi(x_1)$	$\phi(x_2)$	$\phi(x_n)$
t_0	t_1	t_2	$\cdots t_n$

Since x_j is not computable, then $\langle \phi(x) \rangle$ is meaningless. But we would like to know what the long time average is.

$$\langle \phi(x) \rangle = \lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} \phi(x_j) \qquad \text{— (1)}$$

(replacing time average by ensemble average)

$$\langle \phi(x) \rangle = \int dx\, \rho(x)\, \phi(x) \qquad \text{— (2)}$$

If the system is ergodic, (1) = (2): time average = ensemble average.

<u>Note:-</u> $\rho(x)$ should not vary with evolution, and to attain this the dynamics has to run for a very, very long time; that's why we need long time averages. Also, for this reason we need the Liapunov exponent for a very long time.

Non-linear maps

Simplest is $\rightarrow$ Bernoulli map (shift).

<u>Def^n:-</u> (Bernoulli map is the map of the unit interval to itself.)

x_0 lies in $[0,1]$; if x_1 is $f(x_0)$, x_1 lies in $[0,1]$.

$$x_0 \to x_1 \to x_2 \to \cdots \to x_n, \qquad x_0 \in [0,1], \quad x_1 \in [0,1] \implies x_n \in [0,1]$$

<u>Ex:-</u>

$$x_{n+1} = 2x_n \bmod 1$$

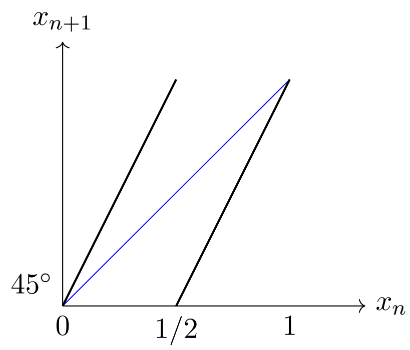
In a map $f(x)$, the fixed point x^* satisfies

$$x^* = f(x^*)$$

Fixed points are given by the intersection of x_{n+1} with the 45° line, which are $\{0,1\}$.

$$f'(0) = f'(1) = 2 > 1$$

So, $x^* = 0, 1$ are unstable fixed points.

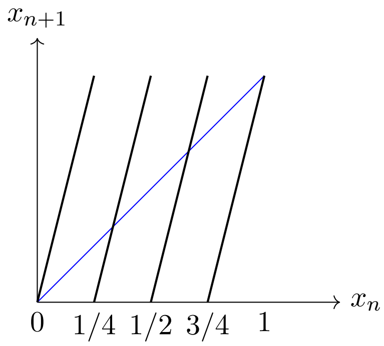


This can also happen: $f(a) = b$, $f(b) = a$.

First iterate of the above map:

$$f^{(2)}(x) = 4x \bmod 1$$

now we have four stable... [fixed] points $\{0, 1/4, 1/2, 3/4, 1\}$; slope at each point is 4, which is greater than 1, so fixed points are unstable.



Hence it seems like the slope is increasing for iterated maps, and there are no stable fixed points.

For $x_0 = 1/3,\ f(x_0) = 2/3,\ f^2(x_0) = 4/3 \bmod 1 = 1/3$: period 2 cycle.

$$\begin{tikzcd} x_0 = 1/3 \arrow[r, bend left] & f(x_0) = 2/3 \arrow[l, bend left] \end{tikzcd}$$

For $x_0 = 1/5$: $x_1 = 2/5,\ x_2 = 4/5,\ x_3 = 8/5 \bmod 1 = 3/5,\ x_4 = 6/5 \bmod 1 = 1/5$: period 5 cycle.

$$\begin{tikzcd} & x_1 = 2/5 \arrow[r] & x_2 = 4/5 \arrow[d] \\ x_0 = 1/5 \arrow[ur] & & \\ & x_4 = 6/5 \bmod 1 = 1/5 \arrow[ul] & x_3 = 8/5 \bmod 1 = 3/5 \end{tikzcd}$$

Any rational number is going to be part of a periodic orbit. So it is clear we have an infinite number of unstable periodic orbits as well.

Note:- Let us write $x_0, x_1, \ldots, x_n$ in binary to see why it is called the Bernoulli shift. $x_0 \in [0, 1]$.

$$x_0 = 0.\, a_0\, a_1\, a_2\, a_3\, \cdots, \qquad a_i = \begin{cases} 0 \\ 1 \end{cases}$$

$$x_0 = \frac{a_0}{2^1} + \frac{a_1}{2^2} + \frac{a_3}{2^3} + \cdots$$

for $x_{n+1} = 2x_n \bmod 1$:

$$x_1 = a_0.\, a_1\, a_2\, a_3 \cdots$$

if $a_0 = 0$, no problem $\implies x_0 < 1/2$

$a_0 = 1 \implies x_0 \in [1/2, 1] \implies (1.\, a_1 a_2 a_3 \cdots) \bmod 1 = 0.\, a_1 a_2 a_3 \cdots$

So, $x_1 = 0.\, a_1\, a_2\, a_3 \cdots$

$\implies$ we are losing information in the forward direction, so we can't invert the map, as the map function is non-linear. (Linear maps are invertible.)

Non-linearity leads to chaos

So it's the non-linearity which leads to chaos.

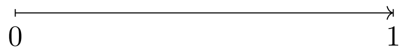
"Any rational number is part of a periodic orbit."

<u>Ex</u>:

$$x_0 = 1/3 = 0.010101\ldots$$
$$x_1 = 2/3 = 0.101010\ldots$$
$$x_2 = 1/3 = 0.010101\ldots = x_0 \implies \text{period 2 cycle}$$

(repetitive in pattern, or terminating.)

Because every point is unstable, every rational number is unstable; it should act like a separatrix which throws out numbers on each side.



On the line from 0 to 1 there are so many more irrational numbers than rational numbers.

And if $x_0 =$ irrational number, it wanders everywhere and fills this interval $[0, 1]$ densely.

→ The system is ergodic.

→ The system is dissipative (How?) → will prove later.

Liapunov exponent for the (1-D) map

$$\lambda(x_0) = \lim_{n\to\infty} \lim_{\epsilon\to 0} \frac{1}{n} \log \frac{|x_n - y_n|}{|x_0 - y_0|}, \qquad |x_0 - y_0| = \epsilon$$
$$= \lim_{n\to\infty} \left(\lim_{\epsilon\to 0} \frac{1}{n} \ln \left| \frac{f^n(x_0+\epsilon) - f^n(x_0)}{x_0 - (x_0+\epsilon)} \right| \right)$$

we get,

$$\lambda(x_0) = \lim_{n\to\infty} \lim_{\epsilon\to 0} \frac{1}{n} \ln \left| \frac{df^{(n)}(x)}{dx} \right|_{x=x_0} \longrightarrow \text{ derivative of } f(x) \text{ at } x = x_0$$

we get,

$$\lambda(x_0) = \lim_{n\to\infty} \lim_{\epsilon\to 0} \frac{1}{n} \ln \left(\frac{df(x_{n-1})}{dx_{n-1}} \cdot \frac{df(x_{n-2})}{dx_{n-2}} \cdots \frac{df(x_1)}{dx_1} \cdot \frac{df(x_0)}{dx_0} \right)$$

from using

$$\{x_0 \to x_1 \to x_2 \to \cdots x_n, \qquad x_{j+1} = f(x_j)\}$$

we can note that $\lambda(x_0)$ is a function which depends on each point of the orbit, $x_0, x_1, x_2, \ldots, x_{n-1}$.

$$f^n(x): \quad x_n = f(x_{n-1}), \quad x_{n-1} = f(x_{n-2}), \quad \ldots, \quad x_1 = f(x_0)$$

$$\lambda(x_0) = \lim_{n\to\infty} \lim_{\epsilon\to 0} \left(\frac{1}{n}\right) \sum_{j=0}^{n-1} \ln |f'(x_j)|$$

for

$$f(x) = 2x \bmod 1, \qquad f'(x) = 2$$

so,

$$\lambda(x_0) = \lim_{n\to\infty} \left(\frac{1}{n}\right)(n \ln 2) = \ln 2$$

also, for $f(x) = ax$, $0 < a < 1$:

$$\lambda(x_0) = \lim_{n\to\infty} \frac{1}{n} n \ln a = \ln a$$

$$f(x) = ax, \qquad f'(x) = a, \qquad a \in [0,1]$$

$f'(x) < 1 \implies$ stable fixed points. $(0 < a < 1)$

Also, for the Liapunov exponent $\lambda(x_0) = \ln a < 0$, so for a stable fixed point the Liapunov exponent becomes negative (everything falls to one place).

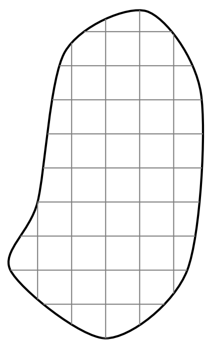
Q.) What would be the Liapunov exponent for a period 2 cycle?

$$f(a) = b, \qquad f(b) = a$$

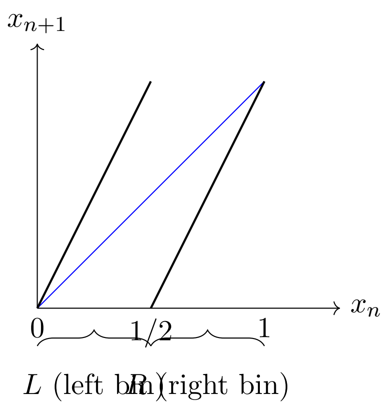
for stability, $|f'(a)f'(b)| < 1$ (slope < 1)

$$\implies \text{Liapunov exponent will be negative for a period 2 cycle.}$$

Note:- The map is random, in the following sense: we agree that the rule is completely deterministic, for each x_0 we have a well-defined x_1 (with no randomness), and yet this is as random as a coin toss. Suppose below is our phase space. In reality we need bins/cells to keep track of phase space points; a 'cell', so we do a partition of phase space into cells/bins and numerically calculate how many times a particular point lies in a particular cell/bin.



For $x_{n+1} = 2x_n \bmod 1$:



We just want to know [whether] a particular point lies in the left bin or the right bin.

Any irrational number from 0 to 1 can be written as non-terminating in binary:

$$x_i = 0.011001111001010111000001001\ldots$$

$$x_0 = 0.\, a_0\, a_1\, a_2\, \cdots\, a_i\, \cdots\, a_n$$

$$x_1 = a_0.\, a_1\, a_2\, a_3\, \cdots\, a_{i+1}\, \cdots\, a_n$$

Above, x_i is equivalent to asking whether a_i is '0' or '1'.

for $x_0 < 1/2$: $a_0 = 0 \leftrightarrow L$

$1 > x_i > 1/2$: $a_0 = 1 \leftrightarrow R$

So any arbitrary string of irrational numbers can be written as

$$LLRLRRLL \cdots LRL \cdots LLR \cdots$$

which can then also be said to be obtained from a coin toss experiment.

So, in this sense the Bernoulli map are as random as coin toss.

So we can't tell the difference if somebody is producing a pattern of **[?]** (L, R, ...) by deterministic rule from the same pattern produced from Bernoulli trial of coin toss.

Statement

For every substring we obtain from the coin toss experiment, there exists an initial condition x_0 which will produce the same string from the deterministic rule of the Bernoulli shift. In that sense randomness and determinism are very closely linked with each other.

Let's find $\rho(x)$:

$$\lambda(x_0) = \lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} \ln \left| f'(x_j) \right|$$

$$\longrightarrow \int_0^1 dx\, \rho(x)\, \ln \left| f'(x) \right|$$

Suppose we start with x_0

$$x_0 \xrightarrow[\text{one time step}]{T} x_1 = f(x_0)$$

What is the probability density of x_1 after one iteration? Of course it is sharp at $x_1 \rightarrow$ Dirac delta function.

$$\rho_1(x_1) = \delta(x_1 - f(x_0))$$

$$\rho_2(x_2) = \delta(x_2 - f(x_1)) \qquad \longrightarrow \text{provided } x_1 \text{ is given}$$

$$= \delta\Big(x_2 - f^{(2)}(x_0)\Big) \qquad \longrightarrow \text{provided } x_0 \text{ is given}$$

$$\rho_2(x_2) = \int \delta(x_2 - f(x_1)) \cdot \delta(x_1 - f(x_0))\, dx_1$$

Moral of the story

any given point $y \xrightarrow[\text{iteration}]{T} f(y)$

after n time steps, what is the probability density in x?

$$\rho_n(x) = \int dy\, \delta(x - f(y))\, \rho_{n-1}(y) \qquad (n = \text{time steps})$$

$$\Big\downarrow \lim(n \to \infty)$$

<u>Frobenius–Perron equation</u> $\qquad \rho(x) = \int dy\, \delta(x - f(y))\, \rho(y) \qquad \to$ invariant density

$\rho_n(x) =$ density of the n^{th} iterate; does not depend on the initial condition.

$\to$ singular integral equation with kernel $K(x, y) = \delta(x - f(y))$

Integral equation

$$\phi(x) = \int dy\, K(x, y)\, \phi(y)$$

$K(x, y) =$ kernel of the integral operator, where the integral operator is $\int dy\, K(x, y)$ acting on $\phi(y)$.

$\to$ Eigenvalue $= 1$

Normalization condition on $\rho(x)$:

$$\int dx\, \rho(x) = 1, \qquad \rho(x) \geq 0,\ x \in I$$

Take $f(y) = 2y \bmod 1$:

$$\rho(x) = \int_0^1 dy\, \delta(x - f(y))\, \rho(y), \qquad f(y) = \begin{cases} 2y & 0 \leq y < 1/2 \\ 2y - 1 & 1/2 \leq y < 1 \end{cases}$$

$$\rho(x) = \int_0^{1/2} dy\, \delta(x - 2y)\, \rho(y) + \int_{1/2}^1 dy\, \delta(x - 2y + 1)\, \rho(y)$$

$$\left\{ \delta(ax) = \frac{1}{|a|} \delta(x) \right\}$$

$$= \left(\frac{1}{2}\right) \int_0^{1/2} dy\, \delta\Big(y - \frac{x}{2}\Big)\, \rho(y) + \frac{1}{2} \int_{1/2}^1 dy\, \delta\left(y - \frac{x+1}{2}\right) \rho(y)$$

$$\rho(x) = \frac{1}{2}\left[\rho\Big(\frac{x}{2}\Big) + \rho\left(\frac{x+1}{2}\right)\right] \qquad \ldots \text{functional equation}$$

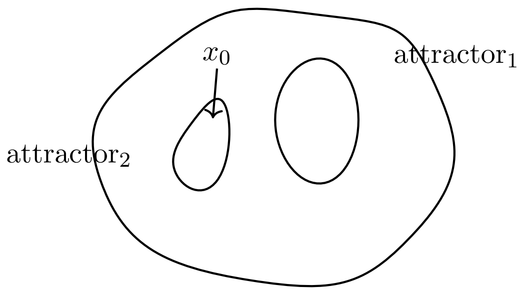
$$\rho(x) = \text{const, from normalizing} \qquad \int_0^1 c\, dx = 1\ (c = 1)$$

$$\boxed{\rho(x) = 1}$$

Therefore iterates of an irrational number uniformly and densely fill up the complete interval.

5 Lecture 17

(Q) Does the Liapunov exponent depend on the initial condition or not?



If the initial condition is such that the trajectory comes to an arbitrary point in phase space, since the Liapunov exponent is calculated by taking time steps to infinity, and the point has explored the whole phase space, therefore in that sense the Liapunov exponent should not depend on the initial condition.

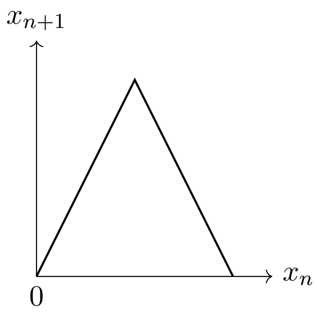
Though there could be multiple attractors, which suggests it depends on x_0, but generally, otherwise, wherever we start, x_0 is going to fill up the whole phase space. So the Liapunov exponent now becomes the property of the trajectory rather than of a point.

Baker's map

(measure (area, volume) is preserved; still the system is chaotic), conservative system.

It is like the Bernoulli map where we have to cut the graph above $x_{n+1} = 1$ and put it back. We could have also folded the piece back rather than cutting, to gain non-linearity. (analogy)

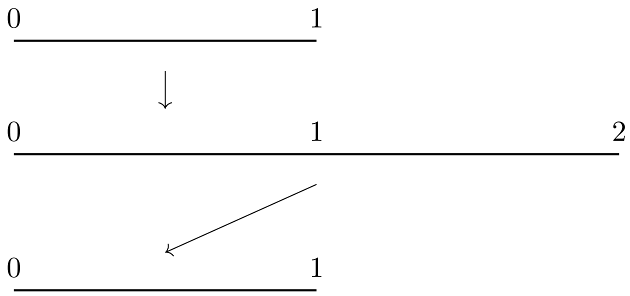
$$x_{n+1} = 2x_n \bmod 1$$



Tent map (non-linear, continuous)

$\lambda(x_0) = \ln 2$, all rational numbers are periodic unstable orbits.

So the idea is to stretch the interval, then put the second half part on top of the first-half map.

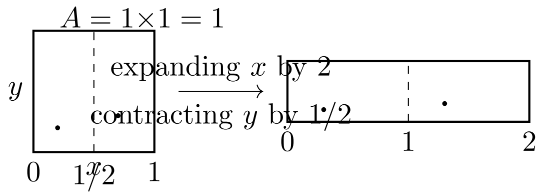


Baker's map takes one more dimension (increasing dimension by one) in such a way that measure is preserved.

$$x_{n+1} = 2x_n \bmod 1 \qquad (\forall)$$

$$y_{n+1} = \begin{cases} \dfrac{1}{2} y_n & (x_n < 1/2) \\ \dfrac{1}{2} + \dfrac{1}{2} y_n & (x_n > 1/2) \end{cases}$$

Let's draw x_{n+1} vs y_{n+1}, and x_n vs y_n.



$$(A = 1{\times}\tfrac{1}{2}) = \tfrac{1}{2}, \text{ folded piece placed on top} \Rightarrow \text{total area } (A_f) = 1$$

- Map is invertible, still chaotic because ($\lambda_1 = \ln 2$)
- Area is preserved ($A_i = 1$, $A_f = 1$)
- $\Rightarrow$ Jacobian of transformation is 1
- Liapunov exponents are $\ln 2,\ \ln\frac{1}{2}$

$$\text{total Liapunov exponent} = \ln 2 + \ln\tfrac{1}{2} = 0$$

As long as measure is preserved, the sum of Liapunov exponents is zero.

All Liapunov exponents are zero for integrable Hamiltonian systems since it is integrable; none of $\lambda(x_0)$ should be +ve, −ve (expanding/contracting volume space), sum of these also will be zero, (otherwise. . .).

For non-integrable Hamilton systems, Liapunov exponents occur in pairs such that the total Liapunov exponent is zero. (−ve for each +ve).

Note: unlike differential dynamics, in discrete time dynamics we can write an explicit solution to a map, but it will still be chaotic.

$$x_{n+1} = 2x_n \bmod 1$$

$$x_n = 2^n x_0 \bmod 1 \qquad \text{— explicit function of time}$$

$\hookrightarrow$ still chaotic.

We can see how we do not see loss of information in Baker's map, if we represent $x_n, y_n, x_{n+1}, y_{n+1}$ in binary:

$$x_n = 0.a_0a_1a_2a_3\cdots$$
$$y_n = 0.b_0b_1b_2b_3\cdots$$
$$\Big\downarrow$$
$$x_{n+1} = 0.a_1a_2a_3\cdots$$
$$y_{n+1} = 0.(a_0b_0)\,b_1b_2b_3\cdots \qquad \text{so we do not lose } a_0.$$

Chaotic <u>but</u> (measure preserving and invertible).

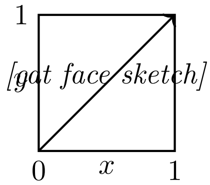
(wow!)

Note: The general rule to produce chaos in a 1-D map is that it has to be non-invertible; for 2 and higher dimensions, invertible maps can also produce chaos.

Arnold's cat map

$$x_{n+1} = (x_n + y_n) \bmod 1$$
$$y_{n+1} = (x_n + 2y_n) \bmod 1$$
$$\begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x_n \\ y_n \end{pmatrix}$$

determinant of transformation matrix = 1 (orientation preserving)



The reason it is named Arnold's cat map is because he put the face of a cat in this graph, and after a few iterations we can't recognize the face of the cat.

$\rightarrow$ every point is having a stretching direction and a contracting direction.

Gauss (continued fraction) map

$$x_0 = \cfrac{1}{a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{a_4 + \cdots}}}}} \xrightarrow{\text{ease of writing}} \frac{1}{a_0}\dot{+}\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \cdots$$

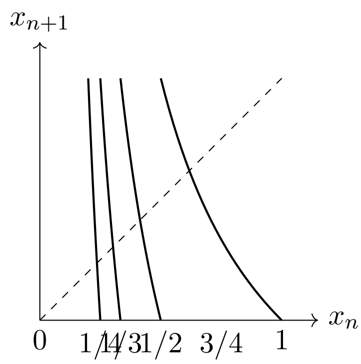
$\hookrightarrow$ notice the '+' sign is written down.

If the continued fraction terminates, x_0 = rational; else x_0 is an irrational number.

$$x_1 = \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{a_4 + \cdots}}}}$$

So to get x_1: take the reciprocal of x_0, throw away the integer part a_0. So,

$$x_{n+1} = \frac{1}{x_n} - \left[\frac{1}{x_n}\right] \qquad \longrightarrow \text{integer part of } (1/x_n)$$



for $\left(\frac{1}{2} < x_n < 1\right)$, $\left(1 < \frac{1}{x_n} < 2\right)$

for any given x_{n+1} there lie infinite pre-images.

slope at each point $|\cdot| > 1 \rightarrow$ unstable at each point.

8.1) Let's take any arbitrary number from 0 to 1. Take its reciprocal, throw away the integer, and repeat the process for a long time. What is the probability that the final number is less than α, $(0, \alpha)$?

$$\left(\text{Ans: } \frac{\ln(1+\alpha)}{\ln 2}\right)$$

Before we calculate, let's find the fixed point of the Gauss map.

f.p$_1 \Rightarrow x = \frac{1}{x} - 1$

$$x^2 + x - 1 = 0 \qquad x = \frac{-1 \pm \sqrt{1+4}}{2} = \frac{-1 \pm \sqrt{5}}{2}$$

So f.p$_{1,2} = \frac{\sqrt{5}-1}{2}$ as $0 < x < 1$

$\rightarrow x + 1 = \frac{1}{x} \;\Rightarrow\; x = \frac{1}{1+x}$

Continued fraction can be written as

$$x = \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cdots}}}$$

For f.point 2, $x = \dfrac{1}{x} - 2$

$$\Rightarrow x^2 + 2x - 1 = 0$$

$$x = \frac{-2 \pm \sqrt{4+4}}{2} = \frac{-2 \pm 2\sqrt{2}}{2} \;\Rightarrow\; -1 \pm \sqrt{2}$$

$$(x = \sqrt{2} - 1)$$

continued fraction $\Rightarrow \dfrac{1}{x} = x + 2$

$$\Rightarrow x = \frac{1}{x+2}$$

$$\Rightarrow x = \cfrac{1}{2 + \cfrac{1}{2 + \cfrac{1}{2 + \cdots}}}$$

Q.) What sort of number is this?

$$x = \cfrac{1}{1 + \cfrac{1}{2 + \cfrac{1}{1 + \cfrac{1}{2 + \cdots}}}}$$

Ans. It would be the fixed point of one iteration of the map, or a period-2 cycle.